PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

skeletal dysplasias

1. Approach to a short child

- **Proportionate short stature:** constitutional, familial, endocrine (hypothyroidism, growth hormone deficiency), nutritional, systemic disease.
- **Disproportionate short stature = skeletal dysplasia** until proved otherwise:
  - **Short-limbed** (achondroplasia, pseudoachondroplasia, metaphyseal dysplasias) vs **short-trunk** (spondyloepiphyseal dysplasias, Morquio).
  - Pattern of limb shortening: **rhizomelic** (proximal: humerus and femur; achondroplasia), **mesomelic** (forearm and leg; Langer, dyschondrosteosis/Leri-Weill with **Madelung deformity**, SHOX gene), **acromelic** (hands and feet).
- Radiographic approach: which part of the bone is affected? **Epiphyseal** (MED, SED), **metaphyseal** (achondroplasia, metaphyseal chondrodysplasias, rickets), **diaphyseal** (Camurati-Engelmann), and is the **spine** involved (spondylo-)?
- **Upper cervical instability screen** (flexion-extension films ± MRI) before anaesthesia in **Morquio, SED congenita, pseudoachondroplasia, Down syndrome, Larsen syndrome, metatropic dysplasia**.

2. FGFR3 group

- **FGFR3 is a negative regulator of chondrocyte proliferation:** activating (gain-of-function) mutations slow growth at the physis.

- **Achondroplasia** (commonest skeletal dysplasia, about 1 in 25,000):
  - **Autosomal dominant**, **FGFR3 G380R** (Gly380Arg) in about 98%; **about 80% are new mutations** associated with **advanced paternal age**. Homozygous achondroplasia is lethal.
  - **Clinical:** rhizomelic shortening, **frontal bossing**, midface hypoplasia, **trident hand** (separation of middle and ring fingers), **genu varum** (fibular overgrowth), thoracolumbar kyphosis in infancy (usually resolves when walking), lumbar hyperlordosis, normal trunk length, normal intelligence. Average adult height about 125-130 cm.
  - **Radiographs: interpedicular distance narrows from L1 to L5** (normally widens), short pedicles, posterior vertebral scalloping, **champagne-glass pelvic inlet**, squared iliac wings, flat acetabular roofs, inverted-V physes at the knee.
  - **Spine problems: foramen magnum stenosis** in infancy (central apnoea, hypotonia, **sudden death**; screen with MRI/polysomnography; suboccipital decompression); **thoracolumbar kyphosis** (avoid unsupported sitting; bracing; fusion if progressive over about 30 degrees after age 3-5); **lumbar spinal stenosis in adults** (neurogenic claudication; wide laminectomy and fusion).
  - **Genu varum:** osteotomy or guided growth if symptomatic. Limb lengthening is controversial.
  - **Vosoritide** (C-type natriuretic peptide analogue; antagonises FGFR3 signalling) increases annual growth velocity.
- **Hypochondroplasia** (FGFR3, milder; normal face). **Thanatophoric dysplasia** (FGFR3; lethal; "telephone-receiver" femora, cloverleaf skull).

3. COMP and type IX collagen group

- **Pseudoachondroplasia** (COMP, cartilage oligomeric matrix protein; AD):
  - **Normal face and normal length at birth**; short limbs become apparent at **2-3 years**.
  - Marked **ligamentous laxity**, windswept knees, **odontoid hypoplasia with C1-C2 instability**, small irregular epiphyses, flared metaphyses, early **osteoarthritis** of hips and knees.
  - Distinguish from achondroplasia: normal face and skull, normal interpedicular distance.
- **Multiple epiphyseal dysplasia (MED)** (COMP, type IX collagen COL9A1-3, matrilin-3; usually AD):
  - Mild short stature, joint pain and stiffness, **small, irregular, flattened epiphyses**, **double-layered patella** (recessive type, SLC26A2).
  - **Bilateral, symmetrical femoral head changes at the same stage**, normal spine, acetabular involvement: differentiates it from **bilateral Perthes** (asynchronous, different stages).
  - **Premature osteoarthritis**: hip arthroplasty in early adulthood.

4. Type II collagen group (COL2A1)

- Type II collagen is in **articular cartilage, physis, vitreous humour and the inner ear**, which explains the eye and hearing problems.
- **Spondyloepiphyseal dysplasia (SED) congenita** (AD, COL2A1):
  - **Short-trunk** dwarfism present at birth, barrel chest, **coxa vara**, **odontoid hypoplasia / os odontoideum with**

atlantoaxial instability (fuse C1-C2 if instability or myelopathy), kyphoscoliosis.
    - **Myopia and retinal detachment** (annual eye examination), hearing loss, cleft palate.
    - Delayed epiphyseal ossification (femoral heads may not be seen for years); early OA.
- **SED tarda:** X-linked (TRAPPC2/SEDL), boys, late childhood, hip and back pain, platyspondyly with a "heaped-up" posterior vertebral body.
- **Kniest dysplasia** (COL2A1): **dumbbell-shaped femora**, large stiff joints, myopia and **retinal detachment**, hearing loss, cleft palate.
- **Stickler syndrome** (COL2A1, COL11A1): myopia and retinal detachment, Pierre Robin sequence, early OA.

5. Other named dysplasias

- **Diastrophic dysplasia** (AR, **SLC26A2** sulfate transporter, common in Finland):
    - **"Hitchhiker" thumb** (short, proximally placed, abducted first metacarpal), **cauliflower ears** (cystic swelling in infancy), cleft palate.
    - **Rigid clubfeet** (very resistant to treatment), joint contractures, **cervical kyphosis** (can cause quadriplegia: monitor and fuse), scoliosis.
- **Cleidocranial dysplasia** (AD, **RUNX2/CBFA1**, master gene for osteoblast differentiation):
    - **Absent or hypoplastic clavicles** (shoulders can be brought together in front of the chest), **delayed closure of fontanelles** and wide sutures with **wormian bones**, **supernumerary teeth** and delayed tooth eruption, **coxa vara**, wide symphysis pubis.

    - Usually needs no treatment; coxa vara with Hilgenreiner-epiphyseal angle over 60 degrees needs valgus osteotomy.
- **Metaphyseal chondrodysplasias:**

Type	Gene / inheritance	Features
Schmid (commonest)	**COL10A1** (type X collagen), AD	Mild short stature, **coxa vara**, genu varum, waddling gait. **Looks like rickets but calcium, phosphate, ALP and vitamin D are normal**
Jansen	**PTH/PTHrP receptor** (activating), AD	Severe, **hypercalcaemia**, hypophosphataemia
McKusick (cartilage-hair hypoplasia)	**RMRP**, AR (Amish)	Fine sparse hair, **immune deficiency**, Hirschsprung disease, risk of lymphoma
Shwachman-Diamond	SBDS, AR	**Pancreatic insufficiency**, neutropenia

- **Mucopolysaccharidoses** (lysosomal storage, defective glycosaminoglycan breakdown; urinary GAGs raised):

Type	Name	Inheritance	Key features
I	**Hurler** (alpha-L-iduronidase)	AR	Coarse face, **intellectual disability**, corneal clouding, gibbus (thoracolumbar kyphosis), carpal tunnel syndrome; stem-cell transplant
II	**Hunter** (iduronate sulfatase)	**X-linked recessive** (the only XLR MPS)	Like Hurler, **no corneal clouding**, milder

Type	Name	Inheritance	Key features
IV	**Morquio** (galactose-6-sulfatase; keratan sulfate)	AR	**Normal intelligence**, short trunk, pectus carinatum, **odontoid hypoplasia with atlantoaxial instability** (occipitocervical fusion often needed), ligament laxity, genu valgum, corneal clouding

    - **Radiographs (dysostosis multiplex):** oar-shaped ribs, bullet-shaped phalanges, anterior beaking of lumbar vertebrae (inferior beaking in Hurler, central in Morquio), proximally pointed metacarpals.
- **Osteopetrosis (marble bone disease): failure of osteoclast resorption**.
    - **Infantile malignant** (AR; TCIRG1, CLCN7): marrow failure (anaemia, hepatosplenomegaly), cranial nerve compression (blindness, deafness), early death without **haematopoietic stem cell transplant** (osteoclasts are of haematopoietic origin).
    - **Adult benign (Albers-Schonberg, AD, CLCN7):** fractures, osteomyelitis of the mandible.
    - **Carbonic anhydrase II deficiency:** with **renal tubular acidosis** and cerebral calcification.
    - **Radiographs:** dense, brittle bones, **bone-in-bone**, **Erlenmeyer flask** distal femora, **rugger-jersey spine**, sandwich vertebrae.
    - Fractures heal slowly; **drilling and fixation are difficult** (very hard bone; use cooled drills and plates rather than nails).
- **Pyknodysostosis** (AR, **cathepsin K**): dense bones, short stature, **acro-osteolysis** of terminal phalanges, open

fontanelles, obtuse mandibular angle, fractures. (Toulouse-Lautrec is thought to have had it.)

- **Camurati-Engelmann** (TGF-beta 1): symmetric **diaphyseal** cortical thickening, leg pain, waddling gait.
- **Melorheostosis:** "dripping candle-wax" cortical hyperostosis along one sclerotome; pain and contractures.
- **Osteopoikilosis** (multiple small sclerotic islands near joints, asymptomatic) and **osteopathia striata** (linear striations): incidental findings.

6. Osteogenesis imperfecta (OI)

- About **90% are COL1A1 or COL1A2 mutations** (type I collagen, quantity or quality), usually AD; recessive forms involve collagen-processing genes (CRTAP, LEPRE1/P3H1, SERPINF1 etc.).
- **Sillence classification:**

Type	Severity	Sclerae	Notes
I	Mild, non-deforming	**Blue**	Commonest; fractures decrease after puberty; **hearing loss** (third decade)
II	**Perinatal lethal**	Blue	Crumpled long bones, beaded ribs
III	**Most severe in survivors**, progressive deformity	Blue at birth, whiter with age	Very short, wheelchair-dependent, **dentinogenesis imperfecta**, triangular face
IV	Moderate	**Normal (white)**	Variable deformity

Type	Severity	Sclerae	Notes
V	Moderate	Normal	**Hyperplastic callus**, **calcification of the interosseous membrane** of the forearm (limits rotation), radial head dislocation; IFITM5; normal collagen

- **Other features: wormian bones** in the skull, ligament laxity, easy bruising, basilar invagination, scoliosis (common, progressive; bracing ineffective; fuse curves over about 45-50 degrees), **dentinogenesis imperfecta**, hyperthermia under anaesthesia.
- **Bisphosphonates** (cyclical IV **pamidronate or zoledronate**) increase bone density and **reduce fracture rate and pain**; they do not cure the collagen defect. **Denosumab** has been tried (rebound hypercalcaemia on stopping).
- **Surgery:** correct long-bone bowing with **multiple osteotomies and intramedullary fixation** (**Sofield-Millar "shish-kebab" osteotomies**). **Telescoping rods** that elongate with growth (**Fassier-Duval**, Bailey-Dubow, Sheffield) reduce reoperation. **Avoid plates** (stress risers, peri-plate fractures).
- **Differential: non-accidental injury.** OI is suggested by blue sclerae, wormian bones, osteopenia, dentinogenesis imperfecta, family history, and fractures in hospital.

7. Neurofibromatosis type 1 (NF1)

- AD, **neurofibromin** (chromosome 17), a tumour suppressor (Ras-GTPase activating protein). Half are new mutations.
- **Diagnostic criteria (2 or more):** 6 or more **cafe-au-lait spots** (over 5 mm before puberty, over 15 mm after), 2 or more

neurofibromas or one plexiform neurofibroma, **axillary or inguinal freckling**, optic glioma, 2 or more **Lisch nodules** (iris hamartomas), a distinctive osseous lesion (**sphenoid wing dysplasia or long-bone cortical thinning / tibial pseudarthrosis**), a first-degree relative with NF1.

- **Orthopaedic problems:**
  - **Dystrophic scoliosis: short, sharp, angular** curves (often kyphoscoliosis), **rib penciling**, vertebral scalloping, enlarged foramina, dural ectasia. **Rapid progression**; bracing fails; early **combined anterior and posterior fusion** (high pseudarthrosis rate). Non-dystrophic curves behave like idiopathic scoliosis but can "modulate" into dystrophic curves.
  - **Congenital pseudarthrosis of the tibia: anterolateral bowing** of the tibia in infancy (about half have NF1). **Protect with a total contact orthosis to prevent fracture**; avoid osteotomy for the bow. Once a pseudarthrosis develops: excision with **intramedullary rodding and bone grafting**, **Ilizarov bone transport**, or **vascularised fibular graft**; bone morphogenetic protein and cross-union techniques. Persistent non-union may end in amputation. (Compare: **posteromedial bowing** is benign and associated with calcaneovalgus foot and leg-length discrepancy; **anteromedial bowing** goes with fibular hemimelia.)
  - **Malignant peripheral nerve sheath tumour** (about 10% lifetime risk; a painful, enlarging plexiform neurofibroma needs MRI and PET).
  - **Hemihypertrophy**, overgrowth (elephantiasis neuromatosa), non-ossifying fibromas.

8. Tumour-like dysplasias

- **Hereditary multiple exostoses (multiple osteochondromas):** AD, **EXT1 (chromosome 8) or EXT2 (chromosome 11)**.
  - Sessile and pedunculated osteochondromas pointing away from the joint; short stature, **forearm deformity** (**ulnar shortening**, radial bowing, radial head dislocation), genu valgum, ankle valgus.
  - **Malignant transformation to secondary chondrosarcoma** in about 1-5% (much higher than a solitary osteochondroma, under 1%). Warning signs: **pain or growth after skeletal maturity**, **cartilage cap over 2 cm** in adults (MRI).
- **Ollier disease** (multiple **enchondromatosis**, unilateral predominance, non-hereditary; IDH1/IDH2 mutations): limb shortening and angular deformity; chondrosarcoma risk about 25-40%.
- **Maffucci syndrome** (enchondromas + **soft-tissue haemangiomas**, phleboliths): higher malignancy risk, including **non-skeletal malignancies** (gliomas, gastrointestinal cancers).
- **Fibrous dysplasia** (**GNAS1** activating mutation, postzygotic): polyostotic forms cause the **shepherd's crook** proximal femur. **McCune-Albright syndrome:** polyostotic fibrous dysplasia, **cafe-au-lait spots with "coast of Maine" borders**, precocious puberty.
- **Fibrodysplasia ossificans progressiva: ACVR1** mutation, **short great toes (hallux valgus) at birth**, progressive heterotopic ossification along fascial planes. **Never biopsy or operate** (flare-ups).

9. Chromosomal and connective tissue conditions with orthopaedic features

- **Down syndrome (trisomy 21): atlantoaxial instability** (atlanto-dens interval over about 5 mm), ligament laxity, **hip instability (dislocation in later childhood)**, **patellofemoral instability**, flatfoot, slipped epiphysis, scoliosis.
- **Marfan syndrome** (FBN1, fibrillin-1; AD): tall, arachnodactyly (**Steinberg thumb sign**, **Walker-Murdoch wrist sign**), **upward lens dislocation**, **aortic root dilatation**, scoliosis (larger, more rigid, higher complication rates), **protrusio acetabuli**, **dural ectasia**, pectus.
- **Homocystinuria** (cystathionine beta-synthase, AR): marfanoid habitus with **downward lens dislocation**, **intellectual disability**, osteoporosis and **thrombosis** (anaesthetic risk).
- **Ehlers-Danlos syndrome:** joint hypermobility, skin hyperextensibility; vascular type (COL3A1) has arterial rupture; avoid stabilisation surgery where possible (poor results).
- **Larsen syndrome** (filamin B): multiple congenital dislocations (knees, hips, elbows), cervical kyphosis, flat face.
- **Turner syndrome:** short stature, **cubitus valgus**, short fourth metacarpal, **Madelung deformity**.

> Exam pearl: **FGFR3 = achondroplasia (interpedicular distance narrows caudally, foramen magnum stenosis). COMP = pseudoachondroplasia and MED. COL2A1 = SED congenita, Kniest, Stickler (eyes!). SLC26A2 = diastrophic (hitchhiker thumb). RUNX2 = cleidocranial. COL10A1 = Schmid. GNAS = fibrous dysplasia. EXT1/2 = HME. ACVR1 = FOP.**

> Exam pearl: Screen the **C1-C2** level before anaesthesia in **Morquio, SED, pseudoachondroplasia and Down syndrome**; watch **cervical kyphosis** in diastrophic dysplasia and Larsen syndrome.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.